

Graph-Assisted Latent MPC: GAS subgoals and cost-to-go inside LeWM’s CEM planner

Working notes and results log

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`scripts/gas_mpc_report.py`)

Current scope (2026-09-17). The active project is graph-assisted planning inside LeWM’s CEM planner, with frozen LeWM dynamics and CEM as the low-level controller. TDR/graph construction, subgoal and cost-to-go objectives, viability or expected-hitting-time terms, and the final-goal switch are the current research mechanisms. Earlier IQL/GCIQL actors, graph-derived auxiliary-value regression, reward shaping, and GAS with a learned low-level policy are historical context, not active development directions. References to those experiments below explain the motivation for the CEM study and preserve their results.

1 Question

LeWM’s planner scores a candidate action sequence by the terminal latent L2 distance $\|z_T - z_g\|^2$ between the predicted endpoint and the goal latent, then executes the CEM optimum receding-horizon (5 blocks of 5 env steps, replanned every 25 steps). It reaches same-episode goals 25 steps ahead 90–94% of the time, 50 steps ahead about 60%, and an arbitrary goal drawn from a *different* episode only 10.8% of the time (250 fixed pairs, 5 seeds; seed 0 alone: 4%). Earlier audits located the failure in the objective: latent L2 stops ranking candidates correctly past ~ 50 steps (Spearman vs. a privileged oracle 0.72 at offset 25, 0.36 at offset 50).

Graph-Assisted Stitching (GAS, Baek et al., ICML 2025) replaces a learned high-level policy with shortest-path search on a graph built in a Temporal Distance Representation (TDR) ψ , where $\|\psi(s) - \psi(g)\| \approx$ the minimum number of steps from s to g . Its subgoals are, by construction, within H_{TD} steps of the current state—inside the range where a local controller’s cost is trustworthy. This document asks whether the same construction, consumed by LeWM’s CEM planner instead of a learned low-level policy, moves the cross-episode and long-horizon numbers. The RL version of GAS on the same latent (see `outputs/gas_pusht`) found the low-level policy, not the graph, was the win; CEM is a much stronger low-level controller (90% vs. 51% at offset 25), so this is a cleaner test of the graph itself.

Answer in one paragraph. The graph subgoal one horizon ahead repairs the mid-range regime on its own (offset 50: 40 $\rightarrow$ 58%) but not arbitrary goals; the viability critic as a feasibility term is what turns the graph’s route into cross-episode successes. Composed naively ($-\log V$, per-population standardisation, goal switch at H_{TD}) the critic lost 8–13 points at offset 25 and needed a different weight β per protocol. Two changes fix both (phase 8, [section 6.5](#)): plan the final horizon straight at the goal with LeWM’s own L2 once the goal is within one lookahead, and replace $-\log V$

by a budget-capped expected hitting time. The resulting single configuration (**OUR**) is at or above every alternative on every protocol: **90.8 / 64.4 / 50.4 / 44.4%** at same-episode offsets 25 / 50 / 100 and cross-episode over five seeds (250 (task, seed) pairs per protocol), against L2’s 91.2 / 48.4 / 11.2 / 10.4 on the same pairs.

2 Setup

Data. All 18,685 Push-T expert episodes (2,336,736 frames), encoded once with the frozen LeWM encoder (CLS latent, 192-d) by `scripts/gas_mpc_prepare.py encode`. Approximately 2% of episodes (373, using integer truncation) are held out from TDR training and default graph construction. Historical evaluation pools draw from the full dataset; the new held-out task convention below uses only these excluded episodes.

TDR. $\psi : \mathbb{R}^{192} \rightarrow \mathbb{R}^{32}$, a 3×512 LayerNorm/GELU MLP on the frozen latent, trained with the HILP/GAS expectile-TD loss (expectile 0.99, $\gamma = 0.99$, goal relabelling 0.625 same-trajectory / 0.375 uniform, two-member target min), 50k steps of batch 1024. Diagnostics: Spearman of $\|\psi(s) - \psi(g)\|$ vs. the privileged state-distance oracle on held-out cross pairs, Spearman vs. the true step gap on same-episode pairs, and the median TDR distance per step gap (the calibration that says what one TDR unit is in env steps).

Graph. Paper Alg. 2 with the official code’s conventions: TE filter at $\theta_{TE} \geq 0.9$, TD-aware sequential clustering at radius $H_{TD}/2$ (GPU implementation, verified bit-identical to the sequential reference), edges between nodes within H_{TD} with weight = TDR distance, goal attached as a temporary node linked to every node within $\max(H_{TD}, 1.2 \cdot d_{\min})$. Each node also carries the latent z of its medoid frame so it can serve as a goal for the L2 cost.

Planner. Unchanged LeWM CEM (300 samples, 30 iterations, top-30, horizon 5 blocks, receding 5 blocks, seeded). Only `model.criterion` is replaced. The criterion sees the current latent z_0 , the five predicted block latents $\hat{z}_1.. \hat{z}_5$ (5, 10, ..., 25 env steps ahead) and the goal latent z_g .

Objectives. With $h(\cdot) = \psi(\cdot)$, $D_g[v]$ the Dijkstra distance from node v to the attached goal, and c_v the node centres:

- **l2:** $\|\hat{z}_5 - z_g\|^2$ (LeWM).
- **tdr:** $\|h(\hat{z}_5) - h(z_g)\|$ (temporal metric, no graph).
- **ctg:** $\min(\|h(\hat{z}_5) - h(z_g)\|, \min_v \|h(\hat{z}_5) - c_v\| + D_g[v])$ (graph cost-to-go as a terminal value; no subgoal commitment).
- **subgoal:** Alg. 1 picks $v^* = \arg \min_{v: \|h(z_0) - c_v\| \leq H_{TD}} D_g[v] + \|h(z_0) - c_v\|$, then walks the shortest path toward the goal until the cumulative TDR distance from z_0 reaches `lookahead`; cost = $\|\hat{z}_5 - z_{\text{med}(v^*)}\|^2$; the true goal once $\|h(z_0) - h(z_g)\| \leq H_{TD}$.
- **subgoal_tdr:** the same node, cost $\|h(\hat{z}_5) - c_{v^*}\|$.
- **dir:** $-\langle h(\hat{z}_5) - h(z_0), \vec{h}_{\text{dir}} \rangle$ with $\vec{h}_{\text{dir}}$ the unit direction to c_{v^*} (the paper’s r^{dir} summed over the horizon); **tdr** once within H_{TD} of the goal.
- **path:** $\sum_{k=1}^5 \|h(\hat{z}_k) - w_k\|$ with waypoints w_k the path nodes at cumulative TDR distance $k \cdot u_5$ from z_0 (u_5 = the calibrated TDR distance of a 5-step gap), the goal once the path ends.

Final-phase switch and critic terms (phase 8). The subgoal objectives switch to the true goal once $\|h(z_0) - h(z_g)\| \leq \theta_{\text{final}}$. The default $\theta_{\text{final}} = H_{\text{TD}}$ (8 units, ≈ 12 steps) still targets a cluster medoid on the first solve of a 25-step task; **final_thresh**=13.7 (one lookahead ≈ 25 steps) plans the whole final horizon at the goal, and **final_metric**=12 scores that phase with LeWM’s $\|\hat{z}_5 - z_g\|^2$ instead of the TDR norm. The viability critic $V(z, z_g, h)$ (outputs/pusht/critic_training/critic_full_sN, with N the seed) enters through one of:

- **nlv**: $-\log V(\hat{z}_5, z_g, h_{\text{rem}})$, h_{rem} the budget left after the plan being scored (**PlanClock**), clipped to the critic’s $h_{\text{max}} = 50$.
- **ET**: $5 \sum_{h \in \{0, 5, \dots, h_{\text{rem}}\}} (1 - V(\hat{z}_5, z_g, h))$, a budget-capped expected hitting time in env steps: feasible endpoints are ordered by how soon they arrive, infeasible ones sit at the cap.

and is composed with the base cost c as either **std**: $\tilde{c} + \beta \tilde{V}$ with $\tilde{x} = (x - \text{med } x) / \text{IQR } x$ over the 300 CEM candidates of each env; **steps**: $\text{steps}(c) + \beta \text{ET}$ with no standardisation, where $\text{steps}(\cdot)$ inverts the empirical median-distance-by-step-gap table (TDR units or z-space L2, gaps 1–50, **gap_calib_s**{seed}.json) so both terms are in env steps and $\beta = 1$ is the nominal weight; or **filter**: rank by c among candidates with $V(\hat{z}_5, z_g, h_{\text{rem}}) \geq \tau$, by V alone when none pass (no weight).

Protocol. Fixed (start, goal) pairs, identical for every objective: outputs/pusht/critic_training/cross_episode (generated from **default_rng**(seed) alone). Same-episode offset K : budget $2K$ (capped at 250). Cross-episode: budget 250. Success = the env’s own **terminated** (agent+block position error < 20 px and block angle error $< \pi/9$). Single seed, 50 pairs per protocol, for screening; promising objectives get more seeds and pairs.

3 Final Results

Headline comparison, LeWM’s own terminal L2 vs. the winning configuration **OUR** (**subgoal_tdr** + final-goal switch at one lookahead + z-space L2 in the final phase + the budget-capped expected-hitting-time critic term, $\beta = 1$; section 6.5), across all three environments and every goal protocol. Each cell is the 5-seed mean $\pm$ sd on the fixed 200-task pool (first 50 tasks used), same pairs for L2 and OUR. Cube’s **same25** column is a hard 22%-pre-solved draw; the representative 200-task correction (section 8.2) gives L2 64.3, OUR 75.5 instead.

Table 1: Push-T. Old-dataset results use the original full-dataset task pool; new-dataset results use **task200u** and planning assets prepared from training episodes only.

configuration	old dataset				new dataset			
	same-ep 25	same-ep 50	same-ep 100	cross-ep	same-ep 25	same-ep 50	same-ep 100	cross-ep
12	91.2 $\pm$ 3.0	48.4 $\pm$ 4.6	11.2 $\pm$ 2.0	10.4 $\pm$ 3.4	91.2 $\pm$ 1.1	40.8 $\pm$ 4.1	13.2 $\pm$ 3.6	14.4 $\pm$ 4.3
OUR	90.8 $\pm$ 2.7	64.4 $\pm$ 4.6	50.4 $\pm$ 5.0	44.4 $\pm$ 3.9	88.4 $\pm$ 2.6	70.0 $\pm$ 2.4	46.4 $\pm$ 3.6	42.4 $\pm$ 3.6

Table 2: Reacher. Old-dataset results use the original full-dataset task pool; new-dataset results use **task200u** and planning assets prepared from training episodes only.

configuration	old dataset				new dataset			
	same-ep 25	same-ep 50	same-ep 100	cross-ep	same-ep 25	same-ep 50	same-ep 100	cross-ep
12	86.8 $\pm$ 4.8	96.0 $\pm$ 1.4	76.0 $\pm$ 1.4	24.4 $\pm$ 1.7	77.2 $\pm$ 5.4	90.8 $\pm$ 1.1	82.8 $\pm$ 1.1	29.2 $\pm$ 2.7
OUR	80.8 $\pm$ 5.2	97.6 $\pm$ 2.6	98.0 $\pm$ 0.0	64.0 $\pm$ 2.8	76.0 $\pm$ 5.1	98.0 $\pm$ 1.4	97.6 $\pm$ 0.9	65.6 $\pm$ 2.2

Table 3: Cube (**same25** is the hard first-50 draw; see caption above for the corrected figure).

configuration	same-ep 25	same-ep 50	same-ep 100	cross-ep
l2	56.8 ± 3.2	50.8 ± 1.6	62.0 ± 2.2	12.8 ± 2.7
OUR	66.8 ± 5.2	56.4 ± 4.6	70.4 ± 0.8	37.6 ± 4.3

Per-seed results

Every cell above, broken down by seed. Pooled = successes summed over all 5 seeds out of 250 (start, goal) pairs per protocol.

Table 4: Push-T, per seed.

protocol	configuration	s0	s1	s2	s3	s4	pooled
same-ep 25	l2	90	92	86	94	94	228/250
same-ep 25	OUR	96	90	90	90	88	227/250
same-ep 50	l2	56	46	50	42	48	121/250
same-ep 50	OUR	60	70	70	62	60	161/250
same-ep 100	l2	10	14	12	12	8	28/250
same-ep 100	OUR	44	52	46	58	52	126/250
cross-ep	l2	12	16	10	8	6	26/250
cross-ep	OUR	48	40	42	42	50	111/250

Table 5: Reacher, per seed.

protocol	configuration	s0	s1	s2	s3	s4	pooled
same-ep 25	l2	80	84	88	92	90	217/250
same-ep 25	OUR	80	76	88	84	76	202/250
same-ep 50	l2	96	96	96	94	98	240/250
same-ep 50	OUR	100	96	98	94	100	244/250
same-ep 100	l2	76	74	76	76	78	190/250
same-ep 100	OUR	98	98	98	98	98	245/250
cross-ep	l2	24	26	24	26	22	61/250
cross-ep	OUR	64	64	60	68	64	160/250

Table 6: Cube, per seed.

protocol	configuration	s0	s1	s2	s3	s4	pooled
same-ep 25	l2	54	52	58	60	60	142/250
same-ep 25	OUR	68	68	74	66	58	167/250
same-ep 50	l2	50	50	50	50	54	127/250
same-ep 50	OUR	52	58	50	62	60	141/250
same-ep 100	l2	62	60	66	60	62	155/250
same-ep 100	OUR	70	70	72	70	70	176/250
cross-ep	l2	16	8	12	14	14	32/250
cross-ep	OUR	34	44	40	38	32	94/250

4 Results

Fixed task set (200 tasks, first 50 used)

All tables in this block: the same 50 tasks for every configuration and every seed; a seed changes only CEM’s random draws. Cells are success rates in %, mean $\pm$ sd over the seeds tested (per-seed counts and values are in the appendix). Default settings unless a row says otherwise: lookahead 13.7 TDR units, $H_{TD} = 8$, TE 0.9, open-loop 25 steps, no retrieval.

Table 7: L2.

variant	same-ep 25	same-ep 50	same-ep 100	cross-ep
default	91.2 $\pm$ 3.0	48.4 $\pm$ 4.6	11.2 $\pm$ 2.0	10.4 $\pm$ 3.4

Table 8: subgoal_tdr.

variant	same-ep 25	same-ep 50	same-ep 100	cross-ep
TE=0.99	84.4 $\pm$ 1.5	54.4 $\pm$ 8.5	21.2 $\pm$ 2.0	18.8 $\pm$ 2.7
default	86.8 $\pm$ 2.7	57.2 $\pm$ 6.3	21.6 $\pm$ 5.7	18.0 $\pm$ 3.3
final@13.7, final L2	90.8 $\pm$ 2.7	61.2 $\pm$ 4.5	28.8 $\pm$ 5.7	19.6 $\pm$ 3.7

Table 9: subgoal_tdr + critic.

variant	same-ep 25	same-ep 50	same-ep 100	cross-ep
final@13.7, final L2, filter $\tau=0.5$	89.3 $\pm$ 2.5	66.7 $\pm$ 6.2	44.0 $\pm$ 4.3	33.3 $\pm$ 4.1
$\beta=0.5$	82.0 $\pm$ 4.2	61.2 $\pm$ 7.2	35.2 $\pm$ 4.1	27.2 $\pm$ 1.6
$\beta=1$	78.8 $\pm$ 6.0	62.4 $\pm$ 4.6	41.6 $\pm$ 7.3	30.8 $\pm$ 3.5
final@13.7, final L2, $\beta=1$, ET(h_{rem})	90.8 $\pm$ 2.7	64.4 $\pm$ 4.6	50.4 $\pm$ 5.0	44.4 $\pm$ 3.9
final@13.7, final L2, $\beta=1$, ET(h_{rem}), steps	90.0 $\pm$ 2.5	62.0 $\pm$ 6.1	36.0 $\pm$ 5.7	32.0 $\pm$ 3.8
retrieval, $\beta=1$	83.6 $\pm$ 2.7	55.6 $\pm$ 2.7	42.8 $\pm$ 3.0	34.4 $\pm$ 5.6
$\beta=2$	74.8 $\pm$ 6.5	57.2 $\pm$ 3.9	41.6 $\pm$ 4.1	34.4 $\pm$ 3.9
final@13.7, final L2, $\beta=2$, ET(h_{rem})	90.0 $\pm$ 1.6	65.3 $\pm$ 4.1	40.0 $\pm$ 0.0	38.0 $\pm$ 3.3
$\beta=4$	68.4 $\pm$ 5.0	58.4 $\pm$ 5.6	37.2 $\pm$ 4.1	34.8 $\pm$ 4.5

Per-seed results

Every test in the blocks above, broken down by seed.

Table 10: Fixed task set (200, first 50), same-ep 25, per seed (CEM repeats); ‘solved’ = tasks solved in ≥ 1 seed / in every seed run.

configuration	s0	s1	s2	s3	s4	mean $\pm$ sd	solved
l2	90	92	86	94	94	91.2 ± 3.0	48 / 40
subgoal_tdr_la13.7_th8_htd8_te0.99	84	84	86	86	82	84.4 ± 1.5	49 / 33
subgoal_tdr_la13.7_th8_htd8_te0.9	86	86	92	86	84	86.8 ± 2.7	50 / 34
subgoal_tdr_la13.7_th8_htd8_te0.9_ft13.7_fl2	90	94	86	92	92	90.8 ± 2.7	49 / 38
subgoal_tdr_la13.7_th8_htd8_te0.9_ft13.7_fl2_ft0.5	86	90	92	–	–	89.3 ± 2.5	49 / 39
subgoal_tdr_la13.7_th8_htd8_te0.9_crit0.5	76	86	86	84	78	82.0 ± 4.2	48 / 33
subgoal_tdr_la13.7_th8_htd8_te0.9_crit1	78	90	76	72	78	78.8 ± 6.0	47 / 28
subgoal_tdr_la13.7_th8_htd8_te0.9_ft13.7_fl2_crit1_ccet	96	90	90	90	88	90.8 ± 2.7	50 / 37
subgoal_tdr_la13.7_th8_htd8_te0.9_ft13.7_fl2_crit1_ccet_steps	94	86	90	90	90	90.0 ± 2.5	50 / 37
subgoal_tdr_la13.7_th8_htd8_te0.9_ret_crit1	84	82	84	88	80	83.6 ± 2.7	48 / 29
subgoal_tdr_la13.7_th8_htd8_te0.9_crit2	70	86	78	72	68	74.8 ± 6.5	50 / 24
subgoal_tdr_la13.7_th8_htd8_te0.9_ft13.7_fl2_crit2_ccet	92	88	90	–	–	90.0 ± 1.6	50 / 38
subgoal_tdr_la13.7_th8_htd8_te0.9_crit4	66	76	66	72	62	68.4 ± 5.0	47 / 19

Table 11: Fixed task set (200, first 50), same-ep 50, per seed (CEM repeats); ‘solved’ = tasks solved in ≥ 1 seed / in every seed run.

configuration	s0	s1	s2	s3	s4	mean $\pm$ sd	solved
l2	56	46	50	42	48	48.4 ± 4.6	36 / 15
subgoal_tdr_la13.7_th8_htd8_te0.99	60	60	60	38	54	54.4 ± 8.5	43 / 5
subgoal_tdr_la13.7_th8_htd8_te0.9	62	48	66	56	54	57.2 ± 6.3	41 / 12
subgoal_tdr_la13.7_th8_htd8_te0.9_ft13.7_fl2	60	60	66	66	54	61.2 ± 4.5	42 / 15
subgoal_tdr_la13.7_th8_htd8_te0.9_ft13.7_fl2_ft0.5	58	72	70	–	–	66.7 ± 6.2	42 / 24
subgoal_tdr_la13.7_th8_htd8_te0.9_crit0.5	64	50	72	62	58	61.2 ± 7.2	45 / 13
subgoal_tdr_la13.7_th8_htd8_te0.9_crit1	62	56	70	60	64	62.4 ± 4.6	45 / 16
subgoal_tdr_la13.7_th8_htd8_te0.9_ft13.7_fl2_crit1_ccet	60	70	70	62	60	64.4 ± 4.6	43 / 18
subgoal_tdr_la13.7_th8_htd8_te0.9_ft13.7_fl2_crit1_ccet_steps	66	56	54	64	70	62.0 ± 6.1	43 / 18
subgoal_tdr_la13.7_th8_htd8_te0.9_ret_crit1	56	52	54	56	60	55.6 ± 2.7	45 / 10
subgoal_tdr_la13.7_th8_htd8_te0.9_crit2	64	56	58	56	52	57.2 ± 3.9	45 / 10
subgoal_tdr_la13.7_th8_htd8_te0.9_ft13.7_fl2_crit2_ccet	60	66	70	–	–	65.3 ± 4.1	46 / 18
subgoal_tdr_la13.7_th8_htd8_te0.9_crit4	66	52	52	60	62	58.4 ± 5.6	42 / 11

Table 12: Fixed task set (200, first 50), same-ep 100, per seed (CEM repeats); ‘solved’ = tasks solved in ≥ 1 seed / in every seed run.

configuration	s0	s1	s2	s3	s4	mean $\pm$ sd	solved
l2	10	14	12	12	8	11.2 ± 2.0	13 / 2
subgoal_tdr_la13.7_th8_htd8_te0.99	20	18	22	22	24	21.2 ± 2.0	24 / 3
subgoal_tdr_la13.7_th8_htd8_te0.9	26	14	20	18	30	21.6 ± 5.7	26 / 2
subgoal_tdr_la13.7_th8_htd8_te0.9_ft13.7_f2	28	26	20	34	36	28.8 ± 5.7	37 / 1
subgoal_tdr_la13.7_th8_htd8_te0.9_ft13.7_f2_ft0.5	38	46	48	–	–	44.0 ± 4.3	38 / 8
subgoal_tdr_la13.7_th8_htd8_te0.9_crit0.5	32	40	40	30	34	35.2 ± 4.1	38 / 4
subgoal_tdr_la13.7_th8_htd8_te0.9_crit1	46	48	30	36	48	41.6 ± 7.3	42 / 3
subgoal_tdr_la13.7_th8_htd8_te0.9_ft13.7_f2_crit1_ccet	44	52	46	58	52	50.4 ± 5.0	43 / 4
subgoal_tdr_la13.7_th8_htd8_te0.9_ft13.7_f2_crit1_ccet_steps	40	34	42	26	38	36.0 ± 5.7	38 / 6
subgoal_tdr_la13.7_th8_htd8_te0.9_ret_crit1	40	40	42	44	48	42.8 ± 3.0	40 / 3
subgoal_tdr_la13.7_th8_htd8_te0.9_crit2	36	44	40	40	48	41.6 ± 4.1	43 / 4
subgoal_tdr_la13.7_th8_htd8_te0.9_ft13.7_f2_crit2_ccet	40	40	40	–	–	40.0 ± 0.0	34 / 7
subgoal_tdr_la13.7_th8_htd8_te0.9_crit4	38	40	36	30	42	37.2 ± 4.1	39 / 1

Table 13: Fixed task set (200, first 50), cross-ep, per seed (CEM repeats); ‘solved’ = tasks solved in ≥ 1 seed / in every seed run.

configuration	s0	s1	s2	s3	s4	mean $\pm$ sd	solved
l2	12	16	10	8	6	10.4 ± 3.4	11 / 2
subgoal_tdr_la13.7_th8_htd8_te0.99	18	22	16	16	22	18.8 ± 2.7	18 / 3
subgoal_tdr_la13.7_th8_htd8_te0.9	22	18	12	20	18	18.0 ± 3.3	17 / 2
subgoal_tdr_la13.7_th8_htd8_te0.9_ft13.7_f2	16	24	18	24	16	19.6 ± 3.7	18 / 4
subgoal_tdr_la13.7_th8_htd8_te0.9_ft13.7_f2_ft0.5	34	38	28	–	–	33.3 ± 4.1	30 / 7
subgoal_tdr_la13.7_th8_htd8_te0.9_crit0.5	26	30	28	26	26	27.2 ± 1.6	28 / 3
subgoal_tdr_la13.7_th8_htd8_te0.9_crit1	24	32	32	32	34	30.8 ± 3.5	31 / 4
subgoal_tdr_la13.7_th8_htd8_te0.9_ft13.7_f2_crit1_ccet	48	40	42	42	50	44.4 ± 3.9	38 / 6
subgoal_tdr_la13.7_th8_htd8_te0.9_ft13.7_f2_crit1_ccet_steps	26	32	36	36	30	32.0 ± 3.8	31 / 4
subgoal_tdr_la13.7_th8_htd8_te0.9_ret_crit1	32	28	42	30	40	34.4 ± 5.6	36 / 4
subgoal_tdr_la13.7_th8_htd8_te0.9_crit2	32	30	38	32	40	34.4 ± 3.9	36 / 6
subgoal_tdr_la13.7_th8_htd8_te0.9_ft13.7_f2_crit2_ccet	34	42	38	–	–	38.0 ± 3.3	28 / 11
subgoal_tdr_la13.7_th8_htd8_te0.9_crit4	30	36	42	36	30	34.8 ± 4.5	34 / 4

5 Diagnostics

Supplementary checks, not part of the headline results above.

5.1 Single-pass L2 CEM at offset 25: predicted latent cost vs. realized latent distance (2026-09-16)

What ran. Every other offset-25 number in this document (e.g. **l2** in the seed-0 screen tables in `legacy_results.tex` and the `same25` protocol) uses budget 50: one CEM solve of 25 actions, executed open-loop, with a second solve available if the env has not terminated by step 25. This diagnostic removes that second chance: budget is capped at exactly 25 env steps, equal to `horizon` $\times$ `action_block` (5×5), so every task gets *exactly one* CEM solve whose full output is executed with no replanning (`scripts/l2_single_pass_diagnostic.py`; verified per-chunk that

the number of CEM solves equals the number of envs in that chunk, i.e. no env ever emptied its action buffer early and forced a second solve). Same fixed 200-task `same_episode` offset-25 set as the rest of this document, single seed (CEM seed 0), plain **12** objective (no graph). Ran on Ada (1×1080 Ti, job 2698367).

Success. **77.0% (154/200)** reach the goal within the single 25-step plan—below the ~90–94% this project sees for offset 25 under the two-solve budget-50 protocol. The gap is exactly the second-chance effect: some tasks whose first CEM solve falls a little short recover on the (here, disallowed) second solve.

Predicted cost vs. realized latent distance. The CEM cost it optimizes, $C^{\text{pred}} = \|\hat{z}_{25} - z_g\|^2$, is computed entirely inside the predictor’s own imagined rollout—it never touches the real dynamics. To measure the resulting model-based planning error directly, the chosen 25-action sequence is executed in the real env, the true final frame is re-encoded through the same frozen LeWM encoder, and the SAME quantity is recomputed against reality: $C^{\text{real}} = \|z_{25} - z_g\|^2$. [Figure 1](#) plots C^{pred} against C^{real} for all 200 tasks, colored by success/failure. Two findings:

- **The predictor is systematically optimistic.** $C^{\text{real}} > C^{\text{pred}}$ on **88.5%** of tasks (every point above the dotted $C^{\text{pred}} = C^{\text{real}}$ line)—CEM is scoring candidates against a rollout that is almost always closer to the goal than what actually happens once the actions are executed in the real env.
- **The realized quantity separates success from failure; the predicted one barely does.** Median C^{real} : **10.9 (success) vs. 30.0 (failure)**, a clear gap. Median C^{pred} : 2.17 (success) vs. 2.31 (failure)—almost no separation. $\text{Spearman}(C^{\text{pred}}, C^{\text{real}}) = 0.297$ ($p = 1.9 \times 10^{-5}$): a real but weak correlation, nowhere near enough for the predicted cost to stand in for the outcome it is meant to approximate.

This is a direct measurement of the model-based planning gap this document’s Question section already infers from the oracle-Spearman numbers (0.72 at offset 25): CEM’s objective is not wrong about direction so much as it is blind to how much its own predictor underestimates real dynamics error, and that blindness is exactly what a plan that only ever sees its own imagined rollout cannot correct for.

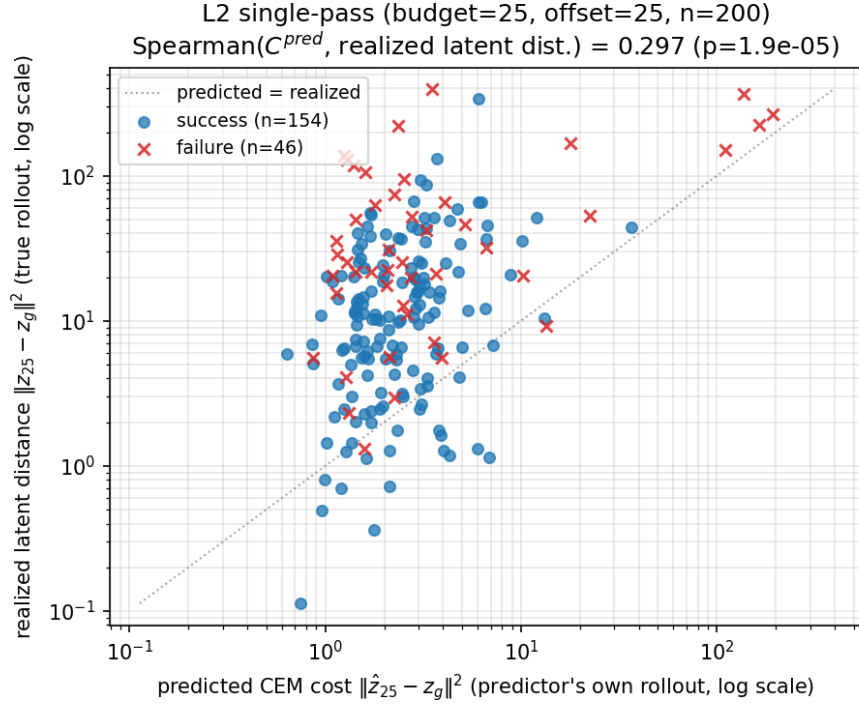


Figure 1: Single-pass ($n = 200$, offset 25, budget 25) L2 CEM: predicted latent cost $\|\hat{z}_{25} - z_g\|^2$ (from the predictor’s own imagined rollout) vs. realized latent distance $\|z_{25} - z_g\|^2$ (true final frame, re-encoded), both log scale, colored by success/failure. Dotted line: $C^{pred} = C^{real}$; points above it are where the predictor was optimistic.

6 Findings log

6.1 Phase 1 (2026-09-14/15): the screen, seed 0, 50 pairs per protocol

What ran. Every objective in the seed-0 screen tables in `legacy_results.tex` on the same three pair sets (same-episode offsets 25 and 50, cross-episode), L2 also at offset 100. L2 and the graph-free TDR cost ran on the laptop (RTX 3050 Ti); the rest on Ada (3×1080 Ti, job 2696687). Lookahead 13.7 TDR units is the calibrated TDR distance of a 25-step gap, i.e. the subgoal is placed one CEM horizon ahead.

Offset 25: no regression, no gain. 78–88% for everything; L2 itself is 88%. The only notable drop is **dir** (78%), the direction-only cost.

Offset 50: the graph helps, and it is the graph, not just the metric. L2 40% → TDR 48% → ctg 50% → subgoal_tdr 58%. The median *final* agent+block error tells the mechanism: L2’s closest approach is 32px but it ends at 76px (it gets near, then the block drifts away); subgoal_tdr’s closest approach is 19px and it ends at 19px. With a subgoal one horizon ahead the planner keeps converging instead of wandering once L2’s ranking goes flat. **path** (5 waypoints) does not help (42%): the waypoint spacing assumes TDR units track env steps block by block, and the intermediate latents are the predictor’s least reliable outputs. **dir** collapses to 26%: an unbounded progress term with no distance in it overshoots the node.

Cross-episode: nothing moves. L2 8%, TDR 10%, ctg 10%, subgoal_tdr 10%, path 6% (4–5 successes out of 50 in every case; the pooled 5-seed L2 baseline is 10.8%). The per-pair diagnostic (the `tab:pair-diag-cross-s0` table in `legacy_results.tex`) is the important number: by the graph’s own geodesic, the 50 cross-episode pairs are **45–86 env steps apart (median 66, max 119)**—every one of them inside the 250-step budget, none of them beyond the horizon the graph objective handles at offset 50. The objective knows a data-supported route; CEM does not execute it. The successes are still the pairs whose block starts near its goal pose (2/8 “block near” vs 3/42 “block far” for subgoal_tdr).

Diagnosis. Two facts together—(i) the graph objective wins at offset 50 by keeping the planner converging near the goal, (ii) it changes nothing on cross-episode pairs that the same graph rates as 45–120 steps away—say the cross-episode bottleneck is the low-level executor, not the objective: 25 actions chosen open-loop from a zero-mean Gaussian proposal, replanned every 25 steps, cannot string together the block pushes that a 66-step expert route needs. That matches the earlier failure analysis of L2 (agent reaches its own target, block displaced but never converging).

Decision. Phase 2 runs both branches: the executor branch—replan every 5–10 steps (**receding** 1/2), warm-start CEM’s mean from the logged 25-action block of the dataset frame that starts nearest the current state and lands nearest the chosen node (**retrieval**), and add the viability critic as a standardised feasibility term (**critic_beta**); (A) confirm the offset-50 gain on seeds 1–2, check cross on seeds 1–2, sweep $H_{TD} \in \{12, 16\}$ and lookahead $\{8, 20\}$, and map offset 100.

6.2 Phase 2 (2026-09-15, Ada): both branches

Branch A – confirmation and sweeps. The offset-50 gain is real across seeds: **subgoal_tdr** 58 / 52 / 56% on seeds 0 / 1 / 2 against L2 40 / 28 / 48% (mean +16.7 points, all three seeds positive, the same 50 pairs per seed for both rows). It also carries to offset 100 (30% vs. L2 22%; tdr and ctg 28%). The sweeps are flat-to-mild: lookahead 8 / 13.7 / 20 TDR units gives 50 / 58 / 56%; $H_{TD} = 12$ (5,632 nodes) gives 64% at offset 50 but 8% on cross-episode, $H_{TD} = 16$ (2,041 nodes) 50%. The z-space variant **subgoal** (LeWM’s own L2 to the node’s medoid frame) is 56% at offset 50 and 16% on cross-episode, so the metric used for the last 25 steps barely matters; *where* the subgoal sits is what matters. Cross-episode with the plain subgoal objective stays at the floor on a second seed too (12% on seed 1).

Executor branch. Three interventions on cross-episode (seed 0, L2 8%, plain subgoal_tdr 10%):

- **Replan every 10 steps (receding 2):** 18% on cross, but 36% at offset 50 (down from 58). Closed-loop tracking helps a little on the hard pairs and hurts the easy ones—the Alg.-1 node re-selection every 10 steps flips targets before the 25-step plan has done its work.
- **Retrieval warm-start** (CEM mean = the logged 25-action block of the dataset frame starting nearest $\psi(z_0)$ and landing nearest the node): 24% on cross (12/50), 54% at offset 50 (no change). Its wins are early (hits at steps 22–71), i.e. it executes the first pushes well, but the final error is unchanged (median 129px)—it does not stop the drift.
- **Viability critic** (cost = standardised subgoal cost + standardised $-\log V(\hat{z}_5, z_g, h_{rem})$, the critic from the viability line, median/IQR over each CEM population): **26% on cross (13/50) and 72% at offset 50 (36/50)**—the best number on both protocols by a wide margin. Its wins are the genuinely far pairs (10 of the 42 “block far” pairs; 9 of the pairs the graph rates 50–100 steps away, vs. 2 for every other objective), its hits spread over the whole

250-step budget, and its median final error is 59 px / 9° (plain subgoal_tdr 125 px / 19°, L2 163 px / 71°). The critic is what stops the drift.

Reading. The subgoal objective fixes *direction* (it keeps the planner converging on a data-supported route one horizon at a time) and the critic fixes *feasibility* (it vetoes endpoints from which the goal is no longer reachable in the remaining budget). Neither alone moves cross-episode; together they triple it. Retrieval is a weaker, complementary lever (better first pushes, no drift control).

Decision. Phase 3: the critic-alone control (L2 + critic, to separate the critic’s contribution from the composition), critic + retrieval, critic + z-space subgoal, critic + $H_{TD} = 12$, $\beta \in \{0.5, 2\}$, offset 100, and seeds 1–4 of the winner with matched L2 rows for a pooled 250-pair comparison against the 10.8% baseline.

6.3 Phases 3–4 (2026-09-15, Ada): controls, combinations, β , five seeds

The pooled result. The `tab:gas-mpc-pooled-cross-perseed-n50` table in `legacy_results.tex`: over the 250 fixed cross-episode pairs (5 seeds $\times$ 50), with L2 re-run in the same harness on the same pairs, **subgoal_tdr + critic reaches 28.0% at $\beta = 1$ (70/250) and 33.6% at $\beta = 2$ (84/250) against L2’s 9.6% (24/250)** (the original 5-seed L2 run gave 10.8%). Every seed improves: $\beta = 2$ gives 38 / 40 / 26 / 34 / 30% against L2’s 8 / 12 / 10 / 8 / 10%. This is the first cross-episode number in this project above the $\sim 10\%$ floor that every RL arm and every plain planning objective has sat at.

Controls: the composition, not the critic alone. L2 + critic on the same seed-0 pairs: 12% at $\beta = 1$, 18% at $\beta = 2$ (vs. 26% and 38% with the subgoal objective underneath); at offset 50, L2 + critic is 50% vs. 72% for the composition. The critic supplies feasibility, the graph subgoal supplies direction; the critic on its own still has L2’s flat long-range ranking to fight. The z-space subgoal composes just as well (subgoal + critic: 26%), so again the node choice matters, not the metric to it. $H_{TD} = 12$ + critic: 22% cross, 56% at offset 50—the coarser graph does not help once the critic is present.

β is the lever, and it is not exhausted. Seed 0 cross-episode: $\beta = 0.5/1/2/4 \rightarrow 14/26/38/44\%$, monotone. At offset 50, $\beta = 1$ and 2 are both 72%; at offset 100, 58% and 52% (L2 22%). The per-pair diagnostic (the `tab:pair-diag-cross-s0` table in `legacy_results.tex`) shows where the extra successes come from: at $\beta = 4$, 17 of the 42 “block far” pairs succeed (L2: 3) and 14 of the pairs the graph rates 50–100 steps away (L2: 2). The median final agent+block error falls from 163 px (L2) to 59 / 35 / 29 px at $\beta = 1/2/4$, and the median block-angle error from 71° to 9–14°: the critic term is what keeps the block from drifting after the first approach. The cost of a larger β is speed—the median first-hit step rises from 96 ($\beta = 1$) to 248 ($\beta = 4$)—so very large β would eventually run into the 250-step budget.

Retrieval on top of the critic. $\beta = 1$ + retrieval warm-start: 44% on seed 0 (22/50, median final error 26 px, 7 of the 8 “block near” pairs and 15 of 42 far ones), the joint best with $\beta = 4$; at offset 50, 64% (critic alone 72%). $\beta = 2$ + retrieval: 30% (below $\beta = 2$ alone, 38%). With one seed these two are within noise of each other (± 3 pairs), so retrieval’s contribution on top of the critic is unresolved; retrieval without the critic (24%) and the critic without the subgoal (12–18%) are both clearly below the composition.

What did not work. Faster replanning (receding 1/2) hurts at offset 50 and only slightly helps cross; combined with retrieval it collapses (6%)—a retrieved 25-step block thrown away after 10 steps is worse than either alone. The direction-only cost (**dir**) and the waypoint-tracking cost (**path**) are below L2 at offset 50. Lookahead and H_{TD} are flat within noise.

Where this leaves the question. GAS-style hierarchy inside latent MPC does two separable things. (1) The graph subgoal one horizon ahead repairs the mid-range regime (offset 50: 40 $\rightarrow$ 58% on three seeds; offset 100: 22 $\rightarrow$ 30%) by giving CEM a target inside the range where its cost still ranks correctly. (2) It does nothing for arbitrary goals on its own, because the executor—25 open-loop actions from a Gaussian proposal—cannot carry out a 45–120-step route even when the graph knows one. Adding the viability critic as a standardised feasibility term is what turns the route into successes: 9.6 $\rightarrow$ 33.6% pooled at $\beta = 2$, 44% on seed 0 at $\beta = 4$. The next experiments are the ones this table points at: $\beta \in \{4, 8\}$ on all five seeds, retrieval + $\beta \geq 2$ on all five seeds, and 100 pairs per seed for the finalists so the ± 3 -pair noise stops deciding rankings.

6.4 Phase 5 (2026-09-15, Ada job 2697130): β and retrieval on all five seeds

What ran. $\beta \in \{4, 8\}$ on every seed, $\beta = 1$ + retrieval on every seed, $\beta = 4$ + retrieval on seeds 0–2, $\beta = 2$ at offsets 50 (seeds 0–1) and 100, $\beta = 4$ at offsets 50 and 100, and a fresh 100-pair cross-episode set (seed 0) for L2, $\beta = 4$, and $\beta = 1$ + retrieval.

The seed-0 extras do not survive. Pooled over 250 pairs (the `tab:gas-mpc-pooled-cross-perseed-n50` table in `legacy.results.tex`): $\beta = 0.5/1/2/4/8 \rightarrow -/28.0/\mathbf{33.6}/27.2/25.6\%$. Seed 0’s 44% at $\beta = 4$ was the top of its noise band (its other seeds: 28/18/22/24%). $\beta = 8$ is clearly past the useful range: on seed 2 it has no success before step 175, i.e. the critic term makes the planner too cautious to move. At offset 50, $\beta = 4$ gives 64% (vs. 72% at $\beta = 1$ and 2) with only 18% of successes inside 50 steps (40% at $\beta = 1$); at offset 100, 50% (vs. 58 / 52%). So $\beta = 2$ is the setting: the best cross-episode number, no cost at offsets 50 or 100.

Retrieval is a wash on top of the critic. $\beta = 1$ + retrieval pooled: 32.4% (22/15/11/25/8 successes per seed—the widest spread of any row); $\beta = 4$ + retrieval on three seeds: 30.0%. Both sit between $\beta = 1$ alone (28.0%) and $\beta = 2$ alone (33.6%), so warm-starting CEM from logged action blocks neither helps nor hurts once the critic is present. On the fresh 100-pair set, L2 is 10%, $\beta = 4$ 36%, $\beta = 1$ + retrieval 36%—consistent with the pooled 50-pair numbers.

Offset 50 is stable across seeds with the critic. $\beta = 2$: 72% on seeds 0 and 1 (plain subgoal 58 / 52%, L2 40 / 28%).

Standing result (superseded by phase 8). `subgoal_tdr` + `critic`, $\beta = 2$: **33.6% over the 250 fixed cross-episode pairs (L2 9.6% on the same pairs and solver), 72% at offset 50 (L2 40%), 52–58% at offset 100 (L2 22%)**. The remaining knobs (retrieval, $\beta > 2$, replan rate, H_{TD} , lookahead) are exhausted at this pair count; the TE-threshold and state-regime (expectile = 0.999, $\theta_{TE} = 0.99$) checks are queued as phases 6–7.

6.5 Phase 8 (2026-09-16, Ada job 2698367): closing the short-range gap and the β dependence

Diagnosis. Three things explain why L2 wins at offset 25 while the critic composition wins everywhere else. (i) The critic is a feasibility classifier: at offset 25 the CEM population is almost entirely feasible, V saturates and $-\log V$ carries only calibration noise; the audit confirms it ranks predictor endpoints worse than L2 exactly there (Spearman vs. oracle 0.60 at $h = 25$ vs. L2’s 0.72; 0.70 vs. 0.36 at offset 50). The β sweep at same25 ($87 \rightarrow 82 \rightarrow 79 \rightarrow 75 \rightarrow 68$) is the planner sliding toward pure- V behaviour, which the viability line measured at 66% closed-loop. (ii) Per-population median/IQR standardisation forces each term to a fixed vote share whatever it knows: a saturated critic term is stretched to unit spread and added with weight β , so β alone decides the regime. (iii) With the final-goal switch at $H_{TD} = 8$ units (≈ 12 steps) the first same25 solve still targets a cluster medoid whose radius ($H_{TD}/2 \approx 6$ steps) is comparable to the success tolerance; neither term has L2’s resolution at the goal (subgoal_tdr alone already loses $91 \rightarrow 87$). Feasibility also has no progress gradient (median first-hit step $96 \rightarrow 248$ as β goes $1 \rightarrow 4$).

What ran. Five configurations on the fixed task set, 50 tasks, seeds 0–2, every protocol (`scripts/ada_gas_mpc_phase8.sh`):

- **A** subgoal_tdr with the final-goal switch at one lookahead (`final_thresh=13.7` units ≈ 25 steps) and LeWM’s z-space L2 as the final-phase cost (`final_metric=l2`): the method reduces to L2 wherever L2 is best.
- **OUR** A + critic with a budget-capped expected hitting time, $ET = 5 \sum_{h \leq h_{rem}} (1 - V(\hat{z}_5, z_g, h))$ in env steps (`critic_cost=et`): ranks feasible endpoints by how soon they arrive, infeasible ones at the cap. Standardised composition, $\beta = 1$; **E** the same at $\beta = 2$.
- **C** A + ET with both terms in env steps and no standardisation (`compose=steps`): the base distance is converted through the empirical median-distance-by-step-gap table (`outputs/gas_mpc/gap_calib.s{seed}.json`, TDR units and z-space L2, 12 gaps from 1 to 50), cost = steps-to-subgoal + ET. A saturated critic then contributes near-zero spread by itself; no β .
- **D** A + hard feasibility filter (`critic_filter=0.5`): rank by the base cost among candidates with $V(\hat{z}_5, z_g, h_{rem}) \geq 0.5$, by V alone if none pass. The “veto” made literal; one calibrated threshold, no weight.

Results (seeds 0–2, 50 tasks; table 9). Every row below is on the same 150 (task, seed) pairs; “old β ” rows are the phase 3–5 configurations restricted to the same three seeds.

configuration	same-ep 25	same-ep 50	same-ep 100	cross-ep
L2	89.3	50.7	12.0	12.7
old subgoal_tdr	88.0	58.7	20.0	17.3
old $\beta = 1$ / $\beta = 2$ ($-\log V$, std)	81.3 / 78.0	62.7 / 59.3	41.3 / 40.0	29.3 / 33.3
A: final switch @13.7 + final L2	90.0	62.0	24.7	19.3
OUR: A + ET, std, $\beta = 1$	92.0	66.7	47.3	43.3
E: A + ET, std, $\beta = 2$	90.0	65.3	40.0	38.0
C: A + ET, steps (no std, no β)	90.0	58.7	38.7	31.3
D: A + filter $\tau = 0.5$ (no β)	89.3	66.7	44.0	33.3

- **(a) is closed by the final-phase switch alone.** Every configuration with `final_thresh=13.7` and z-space L2 in the final phase is within noise of L2 at offset 25 (89–92% vs. 89.3%), including the critic ones that previously lost 8–13 points; median first-hit step is 24 for all of

them, as for L2. The switch also lifts the no-critic subgoal objective at every other protocol ($58.7 \rightarrow 62.0$ at offset 50, $20 \rightarrow 24.7$ at 100), i.e. the medoid-radius resolution floor was costing successes there too.

- **(b) the β dependence is gone with the hitting-time cost.** With ET in place of $-\log V$, $\beta = 1$ is at least as good as $\beta = 2$ on every protocol (92/66.7/47.3/43.3 vs. 90/65.3/40/38); under $-\log V$ the two traded places between same25 and cross. One setting, OUR, is now the best or tied-best row on all four protocols, and it is the best cross-episode number in the project on these seeds (43.3% vs. 33.3% for the previous $\beta = 2$ standing result; final agent+block error 26 px / 9° vs. 40 px / 10° ; median first hit 95 vs. 188 steps). ET keeps a progress gradient among feasible endpoints, which is what $-\log V$ lacked.
- **Same units without standardisation (C) works but is the weakest critic variant.** It matches L2 at offset 25 and beats the old β rows at offset 100 and cross only marginally (38.7 / 31.3). Its final errors (53 px at offset 100, 99 px on cross) say the subgoal term dominates the sum: the calibrated conversion puts a residual of a few TDR units at 5–10 “steps”, comparable to the whole ET range, so the critic’s veto is diluted exactly on the far pairs. The per-population standardisation, whatever its faults at short range, is what gives the critic a full vote at long range; the short-range fault is fixed by the switch, not by the composition.
- **The hard filter (D) is a credible β -free alternative:** 89.3 / 66.7 / 44.0 / 33.3 with one threshold, tied-best at offset 50, and its statistics show the veto doing the intended work (67% of candidates pass at offset 25 and the fallback to V -ranking fires on 3% of solves; 39–41% pass at offset 100 / cross with the fallback on 35–38% of solves). It trails OUR by 10 points on cross because a threshold discards the ranking information among infeasible candidates that ET keeps.

Seeds 3–4 for A, OUR, C (same day, two processes per GPU). Five-seed means (250 (task, seed) pairs per protocol; per-seed values in the appendix tables):

configuration	same-ep 25	same-ep 50	same-ep 100	cross-ep
L2	91.2	48.4	11.2	10.4
old $\beta = 1$ / $\beta = 2$ ($-\log V$, std)	78.8 / 74.8	62.4 / 57.2	41.6 / 41.6	30.8 / 34.4
A: final switch @13.7 + final L2	90.8	61.2	28.8	19.6
OUR: A + ET, std, $\beta = 1$	90.8	64.4	50.4	44.4
C: A + ET, steps (no std, no β)	90.0	62.0	36.0	32.0

The two new seeds move nothing qualitatively and strengthen OUR: its cross-episode per-seed values are 48 / 40 / 42 / 42 / 50 (old $\beta = 2$: 32 / 30 / 38 / 32 / 40; L2: 12 / 16 / 10 / 8 / 6), so every seed is 10–18 points above the previous standing result and OUR is the first configuration to clear 40% on every cross-episode seed. Offset 100 is 44–57 across seeds (old β rows 36–48). At offset 25 OUR is 88–96 against L2’s 86–94, the same band. C stays the weakest critic variant (32.0 cross, 36.0 at offset 100).

Standing result. subgoal_tdr with the final-goal switch at one lookahead, z-space L2 in the final phase, and the budget-capped expected-hitting-time critic term at $\beta = 1$ (OUR): 90.8 / 64.4 / 50.4 / 44.4% at offsets 25 / 50 / 100 / cross over five seeds, against L2’s 91.2 / 48.4 / 11.2 / 10.4 on the same 250 pairs per protocol. Remaining checks: $\beta = 0.5$ under ET, seeds 3–4 for D and E, and the filter (D) at $\tau \in \{0.3, 0.7\}$ before it is dropped.

Two processes per GPU. The seed 3–4 cross-episode runs were the first to share GPUs (two evals per 1080 Ti). Per-run wall time went from 13.7 / 14.9 / 15.8 min (A / OUR / C, one process per GPU, seeds 0–2) to 25.5 / 25.5 / 27.1 min, i.e. 1.7–1.9 $\times$ slower per process for 2 $\times$ the processes: a throughput gain of only 5–20%. The CEM loop keeps a 1080 Ti near 100% on its own, so there is little idle capacity to fill. (A first attempt had pinned all six processes to GPU 0 through a shell bug and ran 5–6 $\times$ slower; those runs were discarded and repeated.)

7 Reacher (dm_control, qpos_match)

Second environment for the two headline objectives: L2 and the phase-8 winner OUR. Pipeline: scripts/ada_gas_mpc_reacher.sh (one resumable script: dataset staging, encoding, per-seed TDR / graph / critic training with one process per GPU, a render-consistency guard, then the evaluation sweep with one process per GPU). Results table: results_table_reacher.tex, regenerated by GAS_MPC_ENV=reacher python scripts/gas_mpc_report.py.

Data and task. quentinll/lewm-reacher: 10,000 episodes of exactly 201 frames (2.01M frames, 224 $\times$ 224), columns qpos/qvel (2 joints each) rather than a `state` column; the state vector used everywhere below is qpos|qvel. The frozen LeWM Reacher checkpoint has the same latent (192-d CLS) and action-block convention (5 actions $\times$ 2 dims = 10) as Push-T. Success is the env’s own `qpos_match` termination: every joint within 0.05 rad of the target qpos, as a *raw* difference (no angle wrap) – the shoulder is an unlimited hinge and the dataset’s shoulder angle is unwrapped over roughly ± 8 rad, so a goal one full turn away counts as far, exactly as the env scores it. Protocols as for Push-T (same-episode offsets 25 / 50 / 100 with budget 2K; cross-episode with budget 200 = the dataset episode length), fixed 200-task sets drawn from the same TASK_SEED, first 50 used, 5 seeds.

Hyperparameters and what was changed for this environment.

- **Unchanged:** TDR (32-d, expectile 0.99, $\gamma = 0.99$, 50k steps), TE threshold 0.9, CEM (300 / 30 / top-30, horizon 5 blocks, open-loop 25 steps), critic architecture and losses ($h_{\max} = 50$, $\delta_{\max} = 60$, 5 members, 60k steps, Bellman + monotonicity), $\beta = 1$, ET cost for OUR.
- **Graph radius and lookahead in env steps, not TDR units.** Push-T’s $H_{TD} = 8$ and lookahead 13.7 are TDR-unit values that happen to equal ≈ 12 and 25 env steps of that TDR. The Reacher TDR has its own scale, so both are read off seed 0’s calibrated median-distance-by-gap table after TDR training (H_{TD} = distance at 12 steps, lookahead = final-goal switch = distance at 25 steps; `calib.json`) and shared by all seeds (Push-T’s per-seed 25-step distances differ by $< 3\%$, so one table is safe and keeps one method tag per configuration).
- **Critic cross-episode negatives gated in joint space.** Push-T only labels a cross-episode pair unreachable when the block poses differ by ≥ 100 px. The Reacher analogue is the joint-space L2 distance; measured on the data, no 50-step window ever moves the joints more than 2.35 rad (99th percentile; 3.81 max), so pairs ≥ 2.5 rad apart (about 60% of random cross pairs, whose median distance is 3.26 rad) are safe negatives and closer ones are left unlabelled.
- **Hindsight labels are tight here.** With the 0.05-rad tolerance the logged arm essentially never passes within tolerance of a later frame before reaching it (on 20k random ($t, t+50$) pairs the first hit is the goal step itself), so the hindsight label is $1[h \geq \delta]$ with no behavioural slack – unlike Push-T’s loose 20-px / 20 $^\circ$ predicate.

Calibration and graph. TDR: Spearman vs. the privileged state-distance oracle 0.93, vs. the true step gap 0.82 (seed 0). The calibrated median-distance-by-gap table (`outputs/gas_mpc_reacher/calib.json`) gives $H_{\text{TD}} = 3.93$ TDR units at a 12-step gap and lookahead/final-switch = 5.78 units at 25 steps – both far smaller than Push-T’s 8 and 13.7, i.e. Reacher’s TDR is a flatter, less spread-out metric. The graph at TE= 0.9 keeps 48% of frames as nodes (≈ 530 – 540 nodes, ≈ 1900 edges, one connected component) – roughly $40\times$ sparser than Push-T’s 20.5k-node graph, consistent with Reacher’s 2-DoF state space needing far fewer landmarks to cover.

Critic. Trained with `viability_train.py --env reacher` on the same architecture as Push-T ($h_{\text{max}} = 50$, $\delta_{\text{max}} = 60$, 5 members, 60k steps, Bellman + monotonicity), cross-episode negatives gated at joint-space $L2 \geq 2.5$ rad (the data never moves more than 2.35 rad, p99, in a 50-step window). Final validation: hindsight AUROC 0.906, ECE 0.010, imagined-rollout AUROC 0.94, monotonicity violation rate 0.4% – in the same range as the Push-T critic’s own diagnostics.

The live env must render what the encoder was trained on. The first evaluation pass on Reacher (2026-09-16/17) gave L2 64.8% at offset 25, twenty points under the LeWM paper’s 86% (Fig. 6; PLDM 78, DINO-WM 79, random 10). The cause was not the planner: `reacher.h5` was rendered with MuJoCo 3.5/3.6 (March 2026), while the Ada environment had MuJoCo 3.12.0, which draws the `dm_control` floor *without* its checker texture. At the same (qpos, qvel) the arm is pixel-identical but the mean pixel difference to the stored frame is 18/255, and the frozen encoder moves the latent by a median of 15.1 units – about a whole 25-step gap (17.4; 1-step 3.0) and not a constant offset (11.8 remains after removing the mean shift). With the arm exactly at the goal the live latent still sits 15.4 from the goal latent, so the terminal-L2 cost never approaches zero; nearest-neighbour state retrieval from live frames into a dataset-frame bank recovers the joint state within 0.05 rad only 3% of the time (dataset frames: 100%). Pinning `mujoco==3.5.0` `dm_control==1.0.37` restores the dataset look exactly (0.0 mean pixel difference) and L2 at offset 25 to 86.8 ± 4.8 (80/84/88/92/90; 84.0 ± 2.4 with the paper’s own 10 CEM iterations instead of our 30) – the paper’s number. The launcher now refuses to evaluate unless three dataset frames re-render to within 4/255 of the stored pixels. Push-T is unaffected (pygame/pymunk, no MuJoCo): the same check gives a 0.49/255 pixel difference and a 0.67-unit latent shift, below its own 1-step gap of 1.06. Every Reacher number in this section is from the corrected render; the first pass (L2 64.8/60.0/63.6/17.2, OUR 68.4/68.4/73.2/36.4) is superseded and its files were deleted. Two protocol differences from the paper remain and are minor: the paper draws Reacher start/goal pairs from a random-policy dataset (`dmc/reacher_random`, not shipped) whereas ours come from the SAC training set, and it uses 10 CEM iterations on Reacher (30 only on Push-T).

7.1 Results (2026-09-17, Ada job 2698921, corrected render)

What ran. The two headline configurations from the Push-T line, on the fixed 200-task set (first 50 used), seeds 0–4, all four protocols: **I2** (LeWM’s own terminal latent L2) and **OUR** (`subgoal_tdr` + final-goal switch at one lookahead (`final_thresh=5.78`) + z-space L2 in the final phase + the budget-capped expected-hitting-time critic term, std composition, $\beta = 1$ – the phase-8 Push-T winner). The phase-3 composition (“old”: $-\log V$, std, $\beta = 1$) was dropped from the default sweep; under the broken render it sat below L2 on every same-episode protocol, as on Push-T, and has not been rerun. No β sweep and no **C/D/E** variants have been run on Reacher. Protocol budgets: same-ep 25/50/100 at $2\times$ the offset (50/100/200 steps); cross-episode at 200 steps (Reacher’s full episode length, vs. Push-T’s 250). All 40 seed $\times$ protocol $\times$ config runs completed in 85 minutes with one process per GPU (the earlier “two per GPU OOMs” was every

process’s ≈ 2 GB EGL render context landing on GPU 0 – dm_control picks EGL device 0 regardless of `CUDA_VISIBLE_DEVICES`; the launcher now sets `MUJOCO_EGL_DEVICE_ID` per slot). Full per-seed detail is in the regenerated appendix, `results_table_reacher.tex`.

configuration	same-ep 25	same-ep 50	same-ep 100	cross-ep
l2	86.8 $\pm$ 4.8	96.0 $\pm$ 1.4	76.0 $\pm$ 1.4	24.4 $\pm$ 1.7
OUR (final switch, final L2, ET, $\beta = 1$)	80.8 $\pm$ 5.2	97.6 $\pm$ 2.6	98.0 $\pm$ 0.0	64.0 $\pm$ 2.8

configuration	s0	s1	s2	s3	s4	pooled
l2 (same25)	80	84	88	92	90	217/250
l2 (same50)	96	96	96	94	98	240/250
l2 (same100)	76	74	76	76	78	190/250
l2 (cross)	24	26	24	26	22	61/250
OUR (same25)	80	76	88	84	76	202/250
OUR (same50)	100	96	98	94	100	244/250
OUR (same100)	98	98	98	98	98	245/250
OUR (cross)	64	64	60	68	64	160/250

OUR’s gain is concentrated where the horizon exceeds one plan. Per-seed OUR–L2: same25 0/−8/0/−8/−14 (OUR loses 6 points pooled, 80.8 vs. 86.8 – the same short-range picture as Push-T, where OUR and L2 tie at 90.8 vs. 91.2), same50 +4/0/+2/0/+2 (tie at the ceiling, 97.6 vs. 96.0), same100 +22/+24/+22/+22/+20 (every seed; OUR solves 49 of the 50 tasks in every seed while L2 stays at 76%), cross +40/+38/+36/+42/+42 (every seed; **2.6**×, 64.0 vs. 24.4% – 160/250 vs. 61/250 hits). Under the broken render the same comparison read as a uniform +4 to +10 with 2× on cross-episode; with frames the encoder recognises, L2 is at the paper’s level at short range and the separation moves entirely to the long-horizon and cross-episode protocols, exactly the regime the subgoal graph and the hitting-time critic are built for.

L2 on Reacher: strong at one or two plans, then it stalls. 86.8 / 96.0 at offsets 25 / 50, but 76.0 at offset 100 and 24.4 cross-episode, against Push-T’s 91.2 / 48.4 / 11.2 / 10.4. Reacher’s 2-DoF joint space keeps the goal “visible” to the terminal-L2 cost far longer than Push-T’s block manipulation does, so the baseline degrades slowly rather than collapsing; the failures at 100 steps and cross-episode are near-misses that stall outside the 0.05-rad band, which is what the graph’s intermediate targets and the critic’s progress term fix.

Caveats. (a) No β sweep, retrieval, or the **C/D/E** ablations from the Push-T phase-8 study have been run on Reacher; $\beta = 1$ is the only value tested. (b) Cross-episode success on Reacher is an intrinsically harder bar than it looks – the shoulder joint is unwrapped ($\approx \pm 8$ rad), so a goal a full turn away scores as far even when the two poses are physically close, which likely caps how high any method’s cross-episode number can go. (c) The short-range loss of OUR (80.8 vs. 86.8 at offset 25) is within one seed’s spread but consistent in sign (0 or negative in every seed); the same final-phase switch that closed this gap on Push-T does not fully close it here, and has not been tuned for Reacher’s TDR scale.

8 Cube (OGBench, swm/0GBCube-v0, single cube)

*Third environment for the two headline objectives, L2 and OUR. Pipeline: `scripts/ada_gas_mpc_cube.sh` (one resumable script, cloned from the Reacher one: dataset download straight onto the compute node’s `/ssd_scratch`, encoding, per-seed TDR / graph / critic with one process per GPU, a render-consistency guard, then the evaluation sweep). Environment hooks: *CubeMechanics* in `scripts/common/envs.py`. Results table: `results-table_cube.tex`, regenerated by `GAS_MPC_ENV=cube python scripts/gas_mpc_report.py`.*

Data and task. quentinll/lewm-cube (cube_single_expert): 10,000 episodes of exactly 201 frames (2.01M frames, 224×224 , 102 GB decompressed – staged only on the compute node, no `/share1` copy). A UR5e arm with a gripper moves one cube to a target position; actions are 5-d (xyz delta, gripper yaw, gripper) in $[-1, 1]$, and the frozen LeWM Cube checkpoint’s action encoder takes 25-d blocks (5 actions $\times$ 5 dims), the same frame-skip-5 convention as the other environments. The h5 stores the full simulator state (qpos 21-d, qvel 20-d) plus flattened privileged / proprio columns; the state vector used by every script below is [block xyz, effector xyz] (6-d, metres), read from `privileged_block_0_pos` and `proprio_effector_pos`. Success is the env’s own predicate (`CubeEnv._compute_successes`, `terminate_at_goal`): the cube within **0.04 m of its target position** – position only, no orientation, and the arm’s own pose is irrelevant. The evaluator restores a task’s start from the dataset row’s qpos/qvel and sets the target from the goal row’s block position and quaternion (`config/eval/cube.yaml` callables). Protocols as for Push-T and Reacher (same-episode offsets 25 / 50 / 100 with budget 2K; cross-episode with budget 200 = the episode length), fixed 200-task sets from the same TASK_SEED, first 50 used, 5 seeds.

Hyperparameters and what was changed for this environment.

- **Unchanged:** TDR (32-d, expectile 0.99, $\gamma = 0.99$, 50k steps), TE threshold 0.9, CEM (300 / 30 / top-30, horizon 5 blocks, open-loop 25 steps), critic architecture and losses ($h_{\max} = 50$, $\delta_{\max} = 60$, 5 members, 60k steps, Bellman + monotonicity), $\beta = 1$, ET cost for OUR. Nothing was tuned on Cube.
- **Graph radius and lookahead calibrated in env steps**, exactly as on Reacher: $H_{TD} =$ seed 0’s median TDR distance at a 12-step gap, lookahead = final-goal switch = the distance at 25 steps (`calib.json`), shared by all seeds.
- **Critic cross-episode negatives gated on block displacement.** The Cube analogue of Push-T’s 100-px block gate is the block-position L2 in metres. Measured on the data, the block moves 0.157 m (median) / 0.311 (p90) / 0.477 (p99) / 0.612 (max) in any 50-step window, so the gate is set automatically to the p99, 0.477 m; pairs whose blocks are farther apart are safe “unreachable within $h_{\max}$ ” negatives, closer ones are left unlabelled.
- **The state-distance oracle is only a diagnostic here.** Raw latent L2 has Spearman -0.04 against z-scored [block, effector] distance (vs. 0.83 against the true step gap), i.e. block-plus-gripper position is not a proxy for latent distance on Cube (the arm’s joint configuration dominates the image). The TDR’s `rho_oracle` column is therefore uninformative for this environment; nothing in training or planning uses it.

Calibration, graph, critic. TDR (seed 0): Spearman vs. the true step gap 0.89 (raw L2 0.83); median distance by gap {1:0.8, 2:1.4, 5:3.4, 10:6.4, 25:14.3, 50:19.4}, giving $H_{TD} = 7.49$ and lookahead / final switch = 14.25 TDR units – almost exactly Push-T’s 8 and 13.7, unlike Reacher’s much flatter 3.9 / 5.8. The graph at TE= 0.9 keeps 75% of frames (1.48M of 1.97M training

frames) and clusters them into 15.8–16.2k nodes with 323–338k edges (mean degree 41), one connected component, ≈ 80 s to build – Push-T scale, not Reacher scale. Critics (5 seeds, 60k steps): held-out hindsight AUROC 0.990–0.991, ECE 0.006–0.009, imagined-rollout AUROC 0.983–0.995, monotonicity violation rate 0.4–0.8%, Bellman residual 0.033–0.037, mean V on gated cross-episode negatives ≤ 0.01 – the tightest critic diagnostics of the three environments (Cube’s hindsight labels are nearly slack-free: the block only moves during the carry segment).

Render check. Following the Reacher lesson, the eval stage first restores three dataset rows into the live env and compares the render to the stored frame (`scripts/gas_mpc_cube_render_check.py`). With `mujoco==3.5.0` and `ogbench==1.2.1` the mean pixel difference is 0.0 / 0.08 / 0.2 (of 255) and the live info state equals the h5 state to the millimetre, so the encoder sees exactly the frames it was trained on. (Practical note: `ogbench` must be pip-installed from the compute node; the venv’s `/usr/bin/python3` is 3.10 there but 3.6 on the login node.)

8.1 Results (2026-09-18, Ada job 2699936)

What ran. **l2** and **OUR** (subgoal_tdr + final-goal switch at one lookahead (`final_thresh=14.25`) + z-space L2 in the final phase + the budget-capped expected-hitting-time critic term, std composition, $\beta = 1$), fixed 200-task sets (first 50 used), seeds 0–4, four protocols; 40 runs in 110 minutes with one process per GPU. No β sweep, retrieval, or **C/D/E** variants. Full per-seed detail in `results_table_cube.tex`.

configuration	same-ep 25	same-ep 50	same-ep 100	cross-ep
l2	56.8 $\pm$ 3.2	50.8 $\pm$ 1.6	62.0 $\pm$ 2.2	12.8 $\pm$ 2.7
OUR (final switch, final L2, ET, $\beta = 1$)	66.8 $\pm$ 5.2	56.4 $\pm$ 4.6	70.4 $\pm$ 0.8	37.6 $\pm$ 4.3

configuration	s0	s1	s2	s3	s4	pooled	solved ≥ 1 / all
l2 (same25)	54	52	58	60	60	142/250	36 / 19
l2 (same50)	50	50	50	50	54	127/250	31 / 19
l2 (same100)	62	60	66	60	62	155/250	37 / 27
l2 (cross)	16	8	12	14	14	32/250	11 / 4
OUR (same25)	68	68	74	66	58	167/250	42 / 24
OUR (same50)	52	58	50	62	60	141/250	34 / 20
OUR (same100)	70	70	72	70	70	176/250	40 / 31
OUR (cross)	34	44	40	38	32	94/250	29 / 9

OUR beats L2 everywhere, including short range. Per-seed OUR–L2: same25 +14/+16/+16/+6/−2, same50 +2/+8/0/+12/+6, same100 +8/+10/+6/+10/+8 (every seed), cross +18/+36/+28/+24/+18 (every seed; **2.9 $\times$** , 37.6 vs. 12.8%, 94/250 vs. 32/250 hits). Nineteen of the twenty seed $\times$ protocol cells favour OUR; the one exception is a 50–50 tie (same50, seed 2). The cross-episode ratio matches Push-T (10.4 $\rightarrow$ 44.4, 4.3 $\times$) and Reacher (24.4 $\rightarrow$ 64.0, 2.6 $\times$). Unlike both, OUR also wins at offset 25 (+10 pooled): on Cube even one plan benefits from the subgoal, because the terminal-L2 cost from a frame where the block has not moved yet is dominated by the arm, not the block (see the oracle diagnostic above), so L2 alone drifts the arm toward the goal *image* rather than the goal *block*.

A large share of the tasks are solved at $t = 0$. Cube’s predicate ignores the arm and the block is stationary for most of an episode (it moves only while carried), so many goal frames have the block exactly where it already is. Tasks whose block starts inside the 4 cm tolerance: same25 22%, same50 16%, same100 **44%**, cross 6%; every such task succeeds at the first step for either method, and the first-hit histograms confirm it (24% / 16% / 44% / 6% of hits at step 1). This is why same100 $\geq$ same25 for both methods – at a 100-step offset the goal frame is often the block sitting at its final position after the carry ended. On the remaining tasks, where the block must actually be moved:

block starts > 4 cm away	same25 ($n=39$)	same50 (42)	same100 (28)	cross (47)
l2	44.6	41.4	32.1	7.2
OUR	57.4	48.1	47.1	33.6

The ordering and the cross-episode gap are unchanged; the absolute numbers are the ones to compare with a manipulation baseline. The headline-table numbers above should be read with this dilution in mind.

8.2 Baseline audit: why L2 same25 reads 56.8 when the LeWM paper reports 74 (2026-09-18)

Script: `scripts/ada_cube_baseline_check.sh` (arms ref, pool, cem10) and `scripts/ada_cube_B_pool.sh`; reference logs under `outputs/gas_mpc_cube/logs_ada/baseline_check/`, our runs under `outputs/gas_mpc_cube/eval/` (`n200` and `_cem10` files).

The published number. LeWM’s Fig. 6 (OGBench-Cube, goal 25 steps ahead, 50-step budget – the same protocol as same25) reports LeWM 74, PLDM 65, DINO-WM 86, GCBC 84, GCIQL 64, GCIVL 56 and **Random 48**. App. D states that CEM runs 30 iterations on Push-T and **10 on the other environments**; the repository’s `config/eval/solver/cem.yaml` (which both `eval.py` and `gas_mpc_eval.py` load) uses 30 everywhere.

No computation mismatch. `gas_mpc_eval.py` $\rightarrow$ `evaluate_pairs` was re-checked line by line against `stable_worldmodel`’s `World.evaluate_from_dataset`, which `eval.py` calls: the same start-row sampling formula (`step  $\leq$  len – offset – 1`, uniform over rows), the same callables (`set_state` ends in `mj_forward`; `set_target_pos` writes `mocap_pos`, which is exactly what `_compute_successes` reads; the arm controller is relative to the live effector pose, so restoring `qpos/qvel` leaves no stale target), the same success signal (`terminated` from `terminate_at_goal`, `success_timing='post'`), the same goal row (`start + offset`), the same action scaler, the same model loading (eval mode, no grad), and the render check of §8. The eval config on Ada is byte-identical to the repository’s. As a direct check, the first 50 tasks of the 200-task runs below reproduce the original 50-task numbers bit for bit (54 / 52 / 58), as they must (same CEM seeds).

The issue is the task draw, plus the paper’s own 50-task noise. Cube success at offset 25 is dominated by how far the block has to move: over the 200-task pool L2 solves 100% of tasks whose block starts within 4 cm of its goal, 81% at 4–8 cm, 55% at 8–12, 53% at 12–16, 38% at 16–20, 36% at 20–30 and 8% beyond 30 cm. Over *all* 1.76M valid offset-25 windows of the dataset, 38.4% are pre-solved (median block move 9.6 cm) – which is also why a random policy scores 48 in the paper. Our fixed 200-task pool drew 31.5% pre-solved (median move 12.0 cm), and the **first-50 prefix** used for every headline number drew only **22%** (median move well above the population’s,

34% of tasks in the 20–30 cm bin vs. 21%). Weighting the per-bin success rates by each set’s move-distance histogram predicts 55.6% for the first 50 (observed 56.8), 61.6% for the pool (observed 64.3) and **65.4% for the population**. The 200-task runs make the prefix effect explicit: tasks 0–49 score 54 / 52 / 58, tasks 50–199 score 64.0 / 67.3 / 71.3.

Reference pipeline, paper protocol verbatim. `python eval.py --config-name=cube policy=lewm-cube eval.num_eval=50 seed=S` (each seed is its own 50-task draw *and* CEM seed, exactly as the paper’s $\pm$ figures are produced):

CEM iterations	s42	s43	s44	s45	s46	mean
30 (repo <code>cem.yaml</code>)	68	82	58	66	74	69.6 ± 8.2
10 (paper App. D)	72	82	64	–	–	72.7

The paper’s 74 is reproduced by the paper’s own setting (10 iterations), and even at 30 iterations the reference mean is 69.6 with single 50-task draws ranging from 58 to 82 (the s43 draw had 40% pre-solved tasks and a median move of 8.7 cm; s44 had 38% / 11.1 cm and scored 58). Our 22%-pre-solved prefix at 56.8 is a low draw of the same distribution, not a different planner: the reference and our evaluator agree once the task mix is controlled (population prediction 65.4 vs. reference-30 mean 69.6, with a ± 8 seed SD).

CEM iterations. 10 vs. 30 iterations on identical tasks and seeds: reference s43 82 / 82, s44 64 / 58, s42 72 / 68; our first-50 tasks 62 / 62 / 58 with 10 iterations vs. 54 / 52 / 58 with 30 (must-move tasks 49.6 vs. 44.6). Fewer iterations help L2 by 3–6 points on Cube – the sharper 30-iteration optimum exploits the weaker Cube predictor – but this is secondary to the task draw. All GAS-MPC runs keep 30 iterations for comparability with Push-T and Reacher.

Corrected baseline (3 seeds, whole 200-task pool).

same25, all 200 tasks, CEM 30	s0	s1	s2	mean	must-move ($n=137$)
l2	61.5	63.5	68.0	64.3 $\pm$ 3.3	47.9
OUR (final switch, final L2, ET, $\beta = 1$)	71.5	76.5	78.5	75.5 $\pm$ 3.6	64.5

These are the numbers to quote for Cube same25. L2 at 64.3 matches the population expectation of the same evaluator (65.4), sits inside the reference pipeline’s seed range at 30 iterations (58–82, mean 69.6) and next to SAGE’s 66.7 for the same frozen checkpoint. OUR keeps its margin on the representative pool: +11.2 pooled (+10/+13/+10.5 per seed), 64.5 vs. 47.9 on the 137 block-must-move tasks, and 77.1 vs. 67.6 on tasks 50–199 alone – the same +10 as on the hard prefix (70.7 vs. 54.7), so the first-50 draw compressed both methods equally and did not manufacture the gap. The other Cube protocols (same50 / same100 / cross) were not re-run on the full pool; their first-50 numbers carry the same caveat (same50’s prefix has 16% pre-solved vs. 18.3% in the population, same100’s 44% vs. 37.6%, so those two are close to representative).

Caveats. (a) **The same25 headline table above uses the first-50 prefix**; read its L2 (56.8) as a hard draw and use the 200-task numbers of this audit for any absolute comparison. (b) No β sweep or phase-8 ablations; $\beta = 1$, ET cost, and the calibrated 12-/25-step radii are the only settings tried. (c) Cross-episode tasks pair a random start with a random goal block position that the expert may never have carried a cube to from that start region; 29 of 50 tasks are solved by OUR in at least one seed but only 9 in all five, so the per-seed cross-episode number is still CEM-noise-limited.

9 Task pools: pre-solved tasks and the task200u convention (2026-09-18)

Scripts: `scripts/gas_mpc_make_tasks.py` (`--audit-old` counts the archived pools), `sample_pairs(reject=...)` in `scripts/viability_cross_episode_baseline.py`, `gas_mpc_eval.POOL`.

What was wrong. The fixed 200-task pools (`task200`, one per environment and protocol, drawn from `TASK_SEED` alone) took the paper’s sampling literally: a uniform dataset row as the start and the row `offset` steps later (or a uniform row of another episode) as the goal, with no other filter. Nothing stops such a pair from being solved before the first action. Whether that happens depends on the environment’s success predicate: Push-T needs agent *and* block within 20 px and the block angle within $\pi/9$, Reacher needs both joints within 0.05 rad, but Cube needs only the block within 4 cm of its target and the block is stationary except while carried. The Cube audit (section 8.2) showed that this both inflates absolute numbers (every such task succeeds at step 1 for any method, including a random policy) and makes them depend on the draw (22% vs. 38% pre-solved between our first-50 prefix and the population).

Pre-solved tasks in the archived task200 pools. Counted with each environment’s own predicate (`EnvMechanics.goal_reached`) on the stored start / goal states, first 50 tasks (the ones every headline table uses) and the whole pool:

environment		same-ep 25	same-ep 50	same-ep 100	same-ep 150	cross-ep
Push-T	first 50	0	0	0	0	0
	all 200	0	0	0	0	0
Reacher	first 50	0	0	0	0	0
	all 200	0	1	1	2	0
Cube	first 50	11 (22%)	8 (16%)	22 (44%)	1 (2%)	3 (6%)
	all 200	63 (31.5%)	32 (16%)	78 (39%)	4 (2%)	5 (2.5%)

Push-T and Reacher results are unaffected (no pre-solved task in any first-50 set; Reacher’s four pool-level hits sit beyond task 50). Every Cube same-episode number in section 8 is diluted by the counts above, and `same100`’s 44% is why it reads higher than `same25` for both methods there. Two-Room has no GAS-MPC pool.

Earlier convention: `task200u`. `sample_pairs` takes `reject=mech.goal_reached` and drops any (start, goal) pair the environment would already accept at $t = 0$. Same-episode pools walk a random permutation of the valid start rows and keep the first 200 unsolved windows (still uniform over unsolved windows, then row-sorted like before); cross-episode pools keep the same 200 start rows and redraw a goal only when it is in the start’s episode or already satisfied. The pools are `pairs_{protocol}_task200u.json` next to the old ones and the run ids carry `task200u`; results record a `pool` field and the report groups by it. The unfiltered `task200` files are untouched (the old sampler is the `reject=None` path and still regenerates them bit for bit), so every archived result stays reproducible. Regenerated on 2026-09-18 for Push-T, Reacher and Cube (15 pools, 0 pre-solved tasks each; the Push-T and Reacher cross-episode pools are identical to the old ones since nothing was rejected there, all same-episode pools are fresh draws). **No evaluation has been run on task200u yet**; the headline tables in section 3 remain `task200` numbers until the sweeps are repeated.

Table 14: Push-T held-out, training-only-asset results: success (%) on 50 fixed tasks per CEM seed. The final column is mean $\pm$ sample standard deviation over five CEM seeds.

protocol	method	s0	s1	s2	s3	s4	mean $\pm$ std
same-ep 25	L2	92	92	90	92	90	91.2 $\pm$ 1.1
	OUR	86	88	90	86	92	88.4 $\pm$ 2.6
same-ep 50	L2	34	40	42	44	44	40.8 $\pm$ 4.1
	OUR	72	72	70	70	66	70.0 $\pm$ 2.4
same-ep 100	L2	18	14	14	12	8	13.2 $\pm$ 3.6
	OUR	48	44	52	44	44	46.4 $\pm$ 3.6
cross-ep	L2	14	14	8	16	20	14.4 $\pm$ 4.3
	OUR	38	42	48	42	42	42.4 $\pm$ 3.6

Current convention: task200u (held-out, disjoint). The label remains `task200u`; its definition now includes episode holdout and non-overlap as well as rejecting trivial tasks. Both start and goal frames are sampled only from the saved TDR asset’s held-out episode split (default asset seed 0; episode split seed 0). The environment’s own success predicate rejects pre-solved pairs before they reserve any frames. Within each same-episode protocol pool, the inclusive trajectory interval $[t_{\text{start}}, t_{\text{goal}}]$ cannot intersect another task’s interval in that episode, including shared endpoints. Cross-episode tasks have distinct start and goal episodes and cannot reuse either endpoint frame within their pool. Different protocol pools are independent: episodes and frames may be reused across protocols.

`gas_mpc_make_tasks.py` writes `pairs_{protocol}_task200u.json`, and `gas_mpc_eval.py` uses the same label in run IDs and the recorded `pool` field. Evaluation rejects a graph whose retained construction frames include any held-out episode. On migration, earlier full-dataset `task200u` files are preserved as `*.full_dataset.json` before their replacement. New run IDs carry an additional `_heldout_disjoint` suffix to avoid reusing earlier evaluation caches, while the pool label stays `task200u`. Historical results remain preserved; existing headline results do not become held-out results by changing the sampler.

Preparation checks and capacity. On the saved Push-T split, sampler checks produced 200 nontrivial, disjoint tasks for each of `same25`, `same50`, `same100`, and `cross`. These are sampler checks, not newly prepared pool files or live control results. The following section records the currently completed held-out-disjoint evaluations; the remaining protocols are still being measured.

10 Held-out task-pool and critic refresh (2026-09-19)

The refreshed Push-T evaluation uses the first 50 tasks from each fixed, held-out-disjoint `task200u` pool and CEM seeds 0–4. All learned planning assets—TDR, graph, and TDR-labelled critic—were prepared from training episodes only; the 373 task-pool episodes are excluded. OUR method is `subgoal.tdr` with graph subgoals, the calibrated 13.53-unit final switch to goal L2, and the standardised budget-capped expected-hitting-time term ($\beta = 1$).

OUR is 29.2 points above L2 at offset 50, 33.2 points above at offset 100, and 28.0 points above on cross-episode goals. At offset 25, L2 remains 2.8 points higher. These results replace the earlier mixed-provenance task-pool/critic matrix.

Table 15: Reacher held-out, training-only-asset results: success (%) on 50 fixed tasks per CEM seed. The final column is mean $\pm$ sample standard deviation over five CEM seeds.

protocol	method	s0	s1	s2	s3	s4	mean $\pm$ std
same-ep 25	L2	74	80	70	84	78	77.2 ± 5.4
	OUR	82	78	68	76	76	76.0 ± 5.1
same-ep 50	L2	92	92	90	90	90	90.8 ± 1.1
	OUR	98	96	98	100	98	98.0 ± 1.4
same-ep 100	L2	82	84	82	84	82	82.8 ± 1.1
	OUR	98	96	98	98	98	97.6 ± 0.9
cross-ep	L2	32	26	28	32	28	29.2 ± 2.7
	OUR	68	66	62	66	66	65.6 ± 2.2

On the new Reacher dataset, OUR is 7.2 points above L2 at offset 50, 14.8 points above at offset 100, and 36.4 points above on cross-episode goals; L2 is 1.2 points higher at offset 25.